

HYDROPHONE CALIBRATION CERTIFICATE

Date of Issue	11 Dec 2025	Certificate No.	20251211-03
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Authorised Signatory

Tricia Dare

Calibrated by

Jayden Neo

Checked by

George Martin

Customer Details

Name

University of Bern

Address

Zentraler Rechnungseingang,
Hochschulstrasse 6,
Bern,
3012,
Switzerland

Instrument Details

Manufacturer

Precision Acoustics

Hydrophone SN

4746

Sensor Type

Needle

Sensor Diameter

0.5 mm

Preamplifier SN

HP34211

DC Coupler SN

DCPS0895

Date Received:

10 Dec 2025

Calibration Conditions

Water Temperature	20 ± 1 °C
Water Treatment	De-ionised, de-gassed, filtered
Water Conductivity	< 5 µS/cm
Soaking Period	> 30 minutes
Acoustic Path Length	375 - 420 mm
Source Signal Type	25 Cycle Toneburst
Transducer Drive Voltage	200 V
Cable Type	RG58, 1.5m
Supply Voltage	230 V
Electrical Impedance	50 Ω

Comments

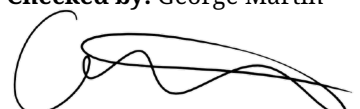
Functional checks were performed prior to the measurements. The equipment was compared against the manufacturer's specification held by the laboratory at the time of calibration. No adjustments were made affecting the calibration. The results apply to the sample as received. This certificate only relates to the items calibrated and is valid only for the configuration stated on this certificate.

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Hydrophone Calibration Procedure

Substitution techniques are well established from one device to another, and the technique used for this calibration is based upon a shocked wave method (Smith and Bacon (1990)). This method achieves a broad calibration bandwidth by utilising the following concepts described in IEC 62127 - part 2 (Annex G): Temporal waveform concept: B Broadband waveform resulting from narrow-band tone-burst utilizing non-linear propagation; Hydrophone position concept: C Far field position with long propagation path to achieve non-linear distortion. In summary, a transducer is driven at sufficiently high power that the resultant acoustic pressure within the water exceeds the limit of linearity of the medium. The subsequent non-linear propagation process causes the shape of the acoustic waveform to change, and significant harmonic content to be generated within the acoustic signal as it propagates. A reference hydrophone that has been previously calibrated at the UK National Physical Laboratory (NPL) is placed at a number of locations within the acoustic field and the waveforms obtained at each of these positions are stored. The waveforms produced by the hydrophone to be calibrated at the same measurement locations are also recorded. The frequency response of the test device can then be calculated from the comparison of the two sets of measurement data. A second reference hydrophone that has been independently calibrated at NPL is then measured, which allows a direct comparison of the two sets of derived sensitivity data to be carried out.

The accuracy of this calibration method is dependent on the ability to accurately place both the reference and test hydrophones at the same position within the acoustic field. Hydrophones are placed at a pre-specified propagation delay away from the source transducer. Measurement positions are then uniquely obtained by moving the hydrophone in the two directions orthogonal to the acoustic axis of the source transducer, in order to maximize the received the hydrophone signal. The hydrophone is rotated about the same two orthogonal axes (centered on the active element of the hydrophone) to correct for any directional response that the hydrophone may exhibit. Maximum sensitivity occurs when the incident waveforms are normally incident upon the active element of the hydrophone. The source transducer has a broad beam profile at its fundamental frequency, but the beam width becomes increasingly narrow at higher harmonic frequencies. This fact is exploited to improve the accuracy of the repositioning operation. A 35 MHz bandpass filter is inserted between the data acquisition unit and the hydrophone, to allow the hydrophone to be aligned to the maximum of the much narrower beam of the 35 MHz harmonic component. The filter is removed once alignment is complete.

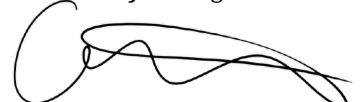
Measurement uncertainty is calculated from the quadrature combination of both Type A and Type B uncertainties. Type A uncertainties are established via statistical methods, and in the case of this calibration are derived from the standard deviation of at least four separate measurements of hydrophone frequency response. Type B uncertainties are obtained via other (non-statistical) methods and in this case includes, but is not limited to, the calibration of the reference hydrophone used in the substitution calibration. The reported expanded uncertainty is based on a standard uncertainty multiplied by a coverage factor $k = 2$, providing a coverage probability of approximately 95%. The uncertainty evaluation has been carried out according to UKAS publication M3003 and the guidelines established in publication JCGM 100: 2008.

- Smith R.A and Bacon D.R. (1990). A Multiple-Frequency Hydrophone Calibration Technique. J. Acoust.Soc.Am. Vol 87 (5), pp. 2231-2243.
- M3003 (2022), The Expression of Uncertainty and Confidence in Measurements. Edition 5, United Kingdom Accreditation Service, London.
- JCGM 100: 2008 Evaluation of measurement data - Guide to the expression of uncertainty in measurement.
- BN EN 62127-2:2007 + A2:2017 Ultrasonic Hydrophones - Calibration for ultrasonic fields up to 40 MHz.

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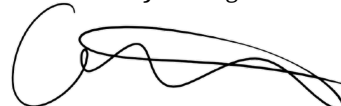
Hydrophone Calibration Data

Frequency (MHz)	Measured Sensitivity (mV/MPa)	Measured Uncertainty (%)
1	659	8.0
2	654	8.0
3	573	8.0
4	492	8.1
5	527	8.1
6	492	8.1
7	525	8.1
8	517	8.1
9	518	8.1
10	532	8.1
11	531	8.1
12	563	8.1
13	577	9.0
14	573	9.0
15	569	9.0
16	568	9.0
17	602	11.7
18	617	11.7
19	623	11.8
20	611	11.8
21	605	12.7
22	610	12.7
23	619	12.7
24	612	12.7
25	572	12.7
26	546	12.7
27	531	12.7
28	497	12.7
29	452	12.7
30	404	12.7

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The Hydrophone Sensitivity vs Frequency graph(s) are visual representations of the 'Hydrophone Calibration Data' table.

Figure 1. Hydrophone Sensitivity (mV/MPa) vs. Frequency (MHz) for system comprising preamplifier SN: HP34211, DC Coupler SN: DCPS0895 and 0.5 mm Needle hydrophone SN: 4746

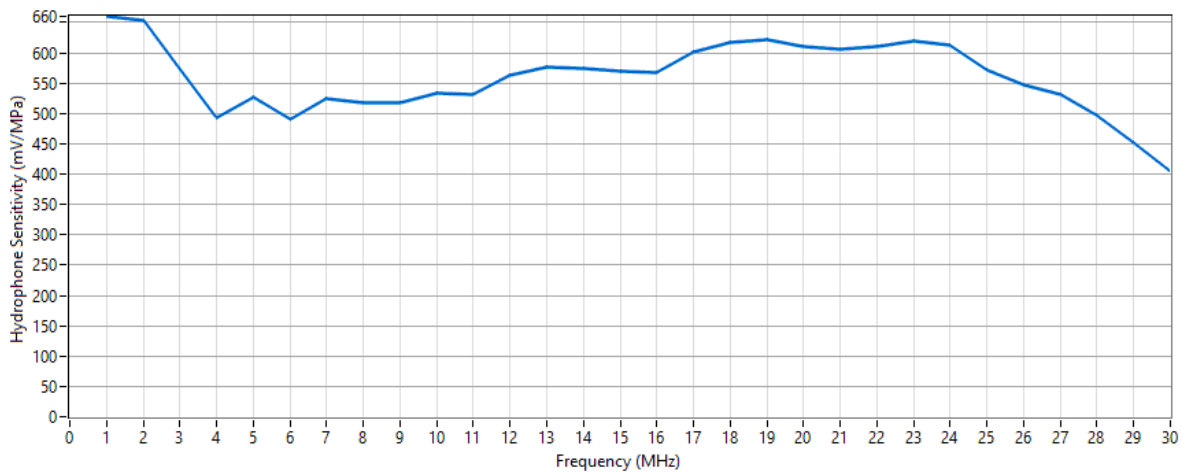


Figure 2. Voltage Waveform (V) vs. Time (s) for system comprising preamplifier SN: HP34211, DC Coupler SN: DCPS0895 and 0.5 mm Needle hydrophone SN: 4746

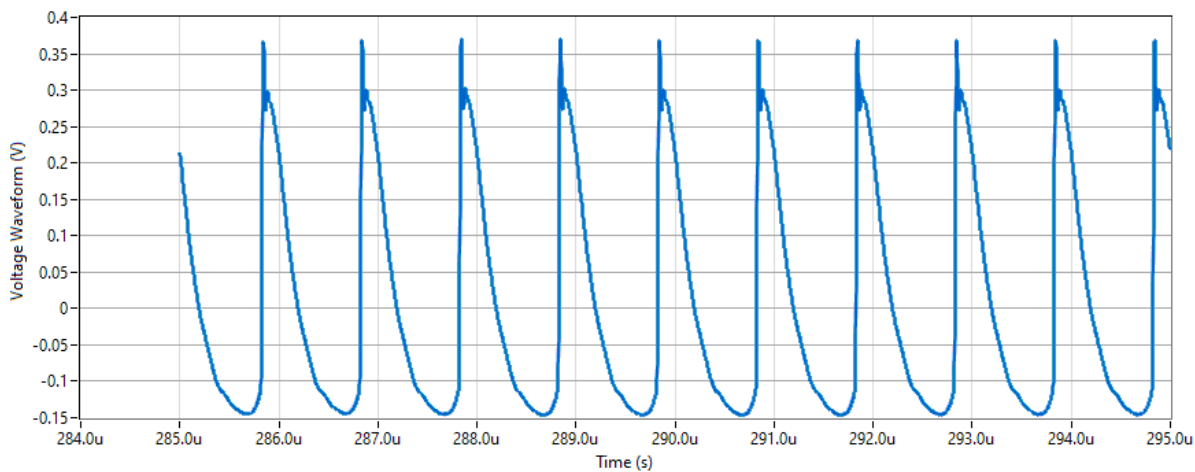
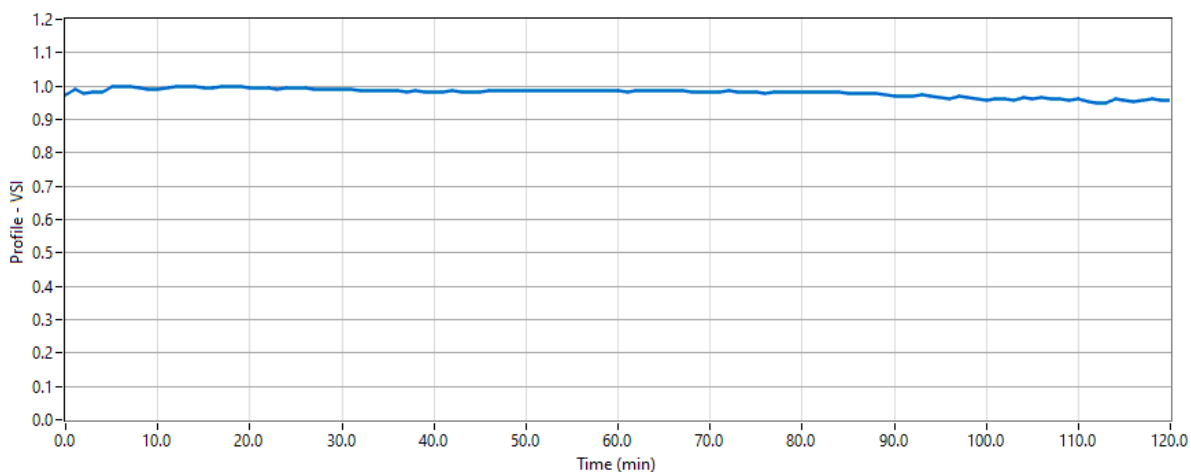


Figure 3. Profile - VSI vs. Time (min) for system comprising preamplifier SN: HP34211, DC Coupler SN: DCPS0895 and 0.5 mm Needle hydrophone SN: 4746



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Traceability Information

Calibration Equipment Used

Device Description	Device S/N	Calibration Lab	Cal. Due Date
Keysight DSO-X-3024A Oscilloscope	MY61410112	UKAS 0452	29 Oct 2027
Type T MI Thermocouple Cable 316 Stainless Steel	223599-A-001	UKAS 0564	01 Jul 2026
PA reference membrane hydrophone	UT1604-323	UKAS 0478	01 Nov 2026

Certificate template ver. 1.4

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